ENVIRONMENTAL PRODUCT DECLARATION

as per *ISO 14025* and *EN 15804+A2*

Owner of the Declaration	Lindner Group
Publisher	Institut Bauen und Umwelt e.V. (IBU)
Programme holder	Institut Bauen und Umwelt e.V. (IBU)
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Valid to	10.05.2031

Lindner Logic 100 Timber Partition, 51 and 52 dB Sound Proofing Variants

Lindner Group

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1. General Information

Lindner Group

Programme holder

IBU – Institut Bauen und Umwelt e.V.
Hegelplatz 1
10117 Berlin
Germany

Declaration number

EPD-LIN-20260204-IBI1-EN

This declaration is based on the product category rules:

Room partition systems, 01.08.2021
(PCR checked and approved by the SVR)

Issue date

11.05.2026

Valid to

10.05.2031

Dipl.-Ing. Hans Peters
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Lindner Logic 100 Timber Partition, 51 and 52 dB Sound Proofing Variants

Owner of the declaration

Lindner Group
Bahnhofstraße 29
94424 Arnstorf
Germany

Declared product / declared unit

The declared unit is 1 m² of Logic 100 Timber Partition, 52 dB, with an 82 mm substructure and weighing 44.51 kg/m².

Scope:

This EPD pertains to the Logic 100 Timber Partition, 51 and 52 dB Sound Proofing Variants, and is a representative EPD, highlighting the product with the highest weighting within its class. The EPD covers four different product variants as listed below:

- Logic 100 Timber Partition, 51 dB, 57 mm substructure
- Logic 100 Timber Partition, 51 dB, 82 mm substructure
- Logic 100 Timber Partition, 52 dB, 57 mm substructure
- **Logic 100 Timber Partition, 52 dB, 82 mm substructure (representative)**

The declared unit is 1 m² of Logic 100 Timber Partition, 52 dB, with an 82 mm substructure and weighing 44.51 kg/m².
The production process of Logic 100 Timber Partitions takes place in two plants, one located in Ostrov, Czech Republic, and the other in Arnstorf, Germany. The production data were collected for the year 2023.
The owner of the declaration shall be liable for the underlying information and evidence; the IBU shall not be liable with respect to manufacturer information, life cycle assessment data and evidences.

The EPD was created according to the specifications of EN 15804+A2. In the following, the standard will be simplified as *EN 15804*.

Verification

The standard EN 15804 serves as the core PCR
Independent verification of the declaration and data according to ISO 14025:2011
☐ internally ☒ externally

Mr Stephen Forson ,
(Independent verifier)

2. Product

2.1 Product description/Product definition

The Lindner Logic 100 Timber Partition is industrially prefabricated with modular components that open up all architectural possibilities. Due to the non-load-bearing partition system, room dimensions can be quickly altered by simply dismantling and reassembling the partition walls in a different configuration.
The product and all its variants consist of three main parts:

1. **Steel System Stud**: Provides structural support, and is made out of steel and required sealing bands and gaskets (TPE and EPDM). Only in the fire protection version of the wall, the systemstud should be covered by fire protection plasterboard in order to provide fire resistance properties. There are two types of steel studs available for all the soundproofing variants of the wall, including sub-structure 57 mm and sub-structure 82 mm. For the fire-protection variants, only sub-structure 82 mm is applicable.
2. **Steel Profiles for floor/ceiling/wall Connection**: Secure the wall to the floor, ceiling, and adjacent walls. The connection profiles are made out of steel, and they require gaskets and sealing tapes (PU-foam band). In order to provide soundproofing properties, they are filled with mineral wool. For the fire protection variants of the wall, the thickness and amount of mineral wool is higher.
3. **Wall Panel**: The main part of the partition wall, made out of wood, forms the visible surface and offers aesthetic benefits. The core wood for all the variants is chipboard. This chipboard can be covered by different surface layers, including HPL veneer, Spruce wood veneer, Beech wood veneer, Oak wood veneer.

The product variants differ based on the materials and components used in each wall type, which depend on the wall's soundproofing or fire protection properties, the type of substructure, and the type of surface coverage. The products covered are available with both substructures (UK 57 and UK 82) and an HPL veneer surface layer.
Since the HPL veneer has a higher weight and a higher environmental impact compared to natural wood veneers (such as Spruce, Beech, and Oak veneer variants), it is assumed that the declared environmental impact for the variants with veneer cover also applies to other variants that only differ in the surface layer and have lower environmental impacts. Hence, the user of the EPD can extend this EPD to other product variants with a natural wood veneer surface cover.

The application and usage of the product is subject to the respective national regulations at the place of use, in Germany, for example, the building regulations of the federal states and the technical provisions based on these regulations.

Logic 100 Timber non-load-bearing partition systems are not subject to CE labeling.

For the use and application of the product the respective national provisions at the place of use apply, in Germany for example the building codes of the federal states and the corresponding national specifications.

2.2 Application

Depending on the product's nature, the system partitions can be utilized in a variety of settings, such as office and conference rooms, industrial workspaces, training and research facilities, as well as in pharmaceutical production, medical technology, microsystems, and precision engineering environments.

The useful service life of non-load-bearing partition walls is ≥ 50 years, according to the ***BBSR Table (code no. 342.411, as of 02/2017)***, published by the Building Institute for Research on Building, Urban Affairs, and Spatial Development. If used properly, there are no costs for maintenance, repair, or replacement during this period.
The partition wall element is designed to be easily disassembled and moved, allowing each wall shell to be removed, relocated, and replaced individually. The product can be maintained and repaired by trained personnel at its place of use, with spare parts provided by the manufacturer throughout its service life.

2.3 Technical Data

The technical data of the products within the scope of the EPD and based on the PCR part B requirements, are listed below.

Constructional data

Name	Value	Unit
Sound absorption for the relevant frequencies 125, 250, 500, 1000, 2000 und 4000 Hz	-	%
Airborne sound reduction	-	dB
Heat transfer coefficient	-	$W/(m^2K)$
Maximum console load acc. to DIN 4103-1	-	kN/m
Maximum horizontal load acc. to DIN 4103-1	-	kN/m
Weight of wall load	-	kN/m^2
Wall Thickness	100, 125, 165	mm
Maximum wall height	5000	mm
Sound insulation	38 - 57	dB

- Logic 100 Timber wall system has a smooth wooden surface. Timber has no significant **absorption coefficient** and is therefore not relevant.
- The wall system is never used as façades, so a **heat transfer coefficient** is not required.

The product is not **harmonised** in accordance with the CPR but in accordance with other provisions for harmonisation of the EU. Performance data of the product according to the harmonised standards, based on provisions for harmonisation:

- ***DIN 4103-1***: Internal non-load bearing partitions requirements and verification, according to the certificate of the structural performance, the horizontal loads of 50 kg/m, 100kg/m and 200kg/m (0.50kN/m, 1.0kN/m and 2.0kN/m) have been taken into account.
- ***DIN EN ISO 10140-2 - Acoustics***: Laboratory measurement of sound insulation of building elements, Part 2:
- ***Measurement of airborne sound insulation DIN EN ISO 10848-2 – Acoustics***: Laboratory and field measurement of flanking transmission for airborne, impact and building service equipment sound between adjoining rooms, Part 2: Application to Type B elements when the junction has a small influence

Logic 100 Timber non-load-bearing partition systems are not subject to CE labelling.

2.4 Delivery status

The products covered by this EPD, **in their delivery status**, are listed below together with their corresponding specifications and grammage.The labeling of the products as Variant 1, Variant 2, etc. is used solely for LCA purposes and data management reasons and does not imply the market name of the product in its delivery status.

Selctive Product Variant	Sound proof / Fire protection propertise	sub-structure	surface coverage	Weight (kg/m2)
Variant 17	SBL Logic Timber 51dB	UK 57	HPL	41,41
Variant 21	SBL Logic Timber 51dB	UK82	HPL	41,54
Variant 25	SBL Logic Timber 52dB	UK 57	HPL	44,39
Variant 29	SBL Logic Timber 52dB	UK82	HPL	44,51

2.5 Base materials/Ancillary materials

The base materials of the Logic 100 Timber Partition include steel for the substructure (system studs) and connection profiles; wood-based materials for the wall panel and its surface layer; plastics and rubber used as sealants; gypsum-based materials providing sound insulation and fire protection properties; and mineral wool.
It should be noted that not all product variants contain all material types. Depending on the required performance properties of the wall system, specific materials may be included or omitted. The table below lists all possible material types for the Logic 100 Timber variants and indicates the material share for each variant where the material is present.

	Material Share (%)			
Product Variant				Representative
Product Weight (kg/m²)	41,41	41,54	44,39	44,51
Material list	Variant 17	Variant 21	Variant 25	Variant 29
Chipboard	58,7%	58,6%	54,8%	54,7%
Metal parts (Steel)	14,0%	14,3%	13,1%	13,3%
Veneer and relevant materials	5,9%	5,9%	5,5%	5,5%
Mineral wool	0,1%	0,1%	0,1%	0,1%
Fire protection add-ons (gypsum fiber board)	20,3%	20,3%	25,7%	25,6%
Rubber-based sealing components	0,8%	0,8%	0,7%	0,7%
Sum	100%	100%	100%	100%

1) 'This product/article/at least one partial article contains substances listed in *the candidate list* (date: 21.01.2025) exceeding 0.1 percentage by mass: no

2) This product/article/at least one partial article contains other carcinogenic, mutagenic, reprotoxic (CMR) substances in categories 1A or 1B which are not on the candidate list, exceeding 0.1 percentage by mass: no

3) Biocide products were added to this construction product or it has been treated with biocide products (this then concerns a treated product as defined by the (EU) *Ordinance on Biocide Products No. 528/2012*): no

2.6 Manufacture

The manufacturing process for the three main parts of the product is as follows:
Steel System Stud:

- The system stud is manufactured in the Czech Republic at the Ostrove production site. The profiles are produced from sheet steel and are kept in stock. They undergo an initial cutting process, and the necessary holes are predrilled to attach the hanging claws.
- The steel floor, ceiling, and wall connection profiles and other steel accessories are manufactured in Germany. These profiles undergo cutting and bending/forming processes to achieve their final shape. If required, both the stud profiles, as well as ceiling and floor profiles, are cut on-site to the appropriate height of the system partition wall to achieve the optimal installation height. Sealing tapes are inserted into the floor and ceiling profiles, while sealings are supplied in rolls and shortened and installed as required.
- The wall panel is processed at the German production site. The chipboard is delivered and cut to size, and may be supplied either with a finished surface or as plain chipboard with the veneer delivered separately. If the veneer is applied to the plain chipboard, the surface is subsequently lacquered, with the type and colour of lacquer being project-specific. After cutting, the edge band is applied.

The above-mentioned components are assembled on-site, where they form a complete system partition wall.

The Lindner Group has a quality management system in accordance with ***EN ISO 9001.***

2.7 Environment and health during manufacturing

The individual components are manufactured in environmentally authorised facilities. The Lindner Group has an energy management system in accordance with ***EN ISO 50001*** and an environmental management system in accordance with ***EN ISO 14001***.
For mineral wool insulation materials, reference must be made to the special regulations in Germany: Ban on the manufacture and use of biopersistent fibres (***Hazardous Substances Ordinance, Annexe IV, No. 22***), Prohibition of placing biopersistent fibres on the market (***Chemicals Prohibition Ordinance, No. 23 of the Annexe to § 1***).

2.8 Product processing/Installation

The individual components of the Logic 100 timber partitions are assembled on the construction site, and no relevant environmental impacts are incurred during the installation of the product. Assembly must be carried out by trained personnel, with the assembly guidelines serving as a reference. Installation is performed using standard tools. All tools and equipment used are subject to operational safety testing in accordance with the ***German Ordinance on Industrial Safety and Health.***

2.9 Packaging

The packaging materials used for packing product components or the final product include reusable wooden pallets made from untreated wood, chipboard, polyethylene (PE)-based plastic materials, and cardboard. For the assessment of reusable wooden pallets, it is assumed that 5 % of the pallets are damaged after use and therefore require disposal and incineration, while 95 % can be reused.

	Material Share (%)			
Product Variant				Representative
Product Weight (kg/m²)	33,49	33,61	33,51	33,64
Material list	Variant 1	Variant 5	Variant 9	Variant 13
Wooden packaging	36,0%	36,1%	36,0%	36,1%
Foam / plastic (PE-based)	34,7%	34,6%	34,7%	34,6%
Paper/cardboard packaging	29,2%	29,1%	29,2%	29,2%
Chipboards	0,1%	0,1%	0,1%	0,1%
Sum	100%	100%	100%	100%

2.10 Condition of use

If used as intended, no material changes in the composition are expected during the use phase.

2.11 Environment and health during use

Under normal use, in accordance with the intended purpose of the system partition wall, no damage or adverse effects on health are expected based on the current state of knowledge.

According to current knowledge, no risks to water, air, or soil are expected if the products described are used as intended.

2.12 Reference service life

The useful life of non-load-bearing walls is ≥ 50 years (according to the ***BBSR table, code no. 342.411, as of 02/2017***, published by the Federal Institute for Research on Building, Urban Affairs and Spatial Development). If used properly, no costs for maintenance, repair, or replacement are expected during this period.

When installed and used as intended for indoor applications, the ageing of the Lindner Logic 100 Timber Partition is not

influenced by continuous mechanical wear due to the static (non-dynamic) nature of the product.
As the system is intended for indoor use and generally does not come into contact with water, no ageing due to moisture is expected under normal conditions; short-term moisture during cleaning does not damage the system, provided it can dry completely.
In the event of mechanical damage, durability and functionality may be impaired, but depending on the extent of damage, affected surfaces/components can be repaired by replacement or reassembly by trained personnel, with spare parts available from the manufacturer throughout the service life.
The partition wall element is easy to relocate. Each wall shell can be removed, moved, and replaced individually. The product can be maintained and repaired by trained personnel at the place of use. Spare parts are provided by the manufacturer throughout the service life of the product.

2.13 Extraordinary effects

Fire
The essential components of the products are non-flammable. Non-essential components, such as seals, are classified as building material class B2.
Information on fire performance is provided in accordance with ***EN 13501-1*** or established national standards. According to***EN 13501-1***:

- The classes of building products with regard to fire performance are defined as A1, A2, B, C, D, E, and F;
- The classes for flaming droplets or particles are defined as d0, d1, and d2;
- The classes for smoke production are defined as s1, s2, and s3.

Fire protection

Name	Value
Building material class	B, D
Burning droplets	s1, s2
Smoke gas development	d0

Water
The Lindner system partition wall is installed indoors and generally does not come into contact with water. Short-term exposure to moisture during cleaning does not damage the system, provided that it is able to dry completely afterwards. Long-term exposure of the partition wall to water may lead to corrosion and the formation of stains on the surfaces of certain components.

Mechanical destruction
In the event of mechanical damage, the durability and functionality of the system may be impaired. Depending on the extent of the damaged surfaces, they can be repaired by replacement or reassembly without impairing overall functionality.

2.14 Re-use phase

Most of the essential and main components of the system partition wall can be dismantled non-destructively and **reused** in an unchanged form for the same application, or even repurposed and reused in a different application.
Some individual components may be damaged during dismantling and therefore need to be newly produced. These components can either be **recycled** or used for **energy recovery**.
For main components such as the wooden parts of the wall panels, if reuse is not intended, they can be directed to **recycling** or **energy recovery** processes.
Depending on the intended end-of-life scenario of the product, the separation of partition wall components for reuse, recycling, energy recovery, or disposal from other building materials is recommended to take place on the construction site to ensure appropriate treatment.

2.15 Disposal

Depending on the intended end-of-life scenario, the system partition wall residues generated on the construction site should be **recycled** wherever possible. If recycling is not feasible, the materials should be directed to **energy recovery**. For certain materials, such as gypsum and mineral wool, for which neither recycling nor energy recovery is possible, disposal in **landfill** is required.

Disposal dismantled

- Wood --> AVV 170201
- Joints/Seals --> AVV 170904
- Mineral wool --> AVV 170604

Disposal in the composite

- Mineral wool --> AVV 170604

2.16 Further information

Additional information about the product can be found at:

https://www.lindner-group.com/de/produkte/wand/systemtrennwaende-vollwand/lindner-logic-100-timber

3. LCA: Calculation rules

3.1 Declared Unit

The declared unit is 1 m² of the **representative product**, Logic 100 Timber Partition, with a sound insulation performance of 49 dB, an 82 mm substructure, and a mass of 33.64 kg/m².

Declared unit and mass reference

Name	Value	Unit
Declared unit	1	m^2
Grammage	44.51	kg/m^2
Layer thickness	0.125	m

The reliability of the LCA results is strongly driven by the quality of the foreground data collected from real life, and production plants, as well as on the underlying background datasets and already explained criteria under sections 7.1 (Foreground data) and 7.4 (technological, time-related, and geographical coverage, plus completeness and precision): the closer the background data match the real production situation in Germany/Czech Republic, the more robust the calculated impacts. In this project following measures are valid regarding the data quality:

- Foreground data is based on the real-world data, all collected directly from the manufacturing process and production sites in Germany and the Czech Republic for the base year 2023.
- Background database is from *the Sphera MLC (fka GaBi) CUP 2023.02*
- Background has been chosen among primary datasets reflecting actual production processes, considering the geographical differences based on the supplier (Means if the material has been sourced from a German supplier, the background data set is DE, if it is a Czech supplier, the background data set is CZ. Same applies for the electricity datasets).

- According to the quality assessment, all of the datasets and processes receive 1 or 2 scores in technological, time-related, and geographical coverage, plus completeness and precision.
- According to the quality assessment, all of the datasets and processes have the reference years close to 2022/2023, which matches the base year of the foreground data.

It has to be noted that, the use of proxy datasets (e.g., HPL as a substitute for veneer, PU-based resins as an adhesive proxy) and 'not evaluated'/older datasets (e.g., 2002/2005/2006) or global datasets instead of regional ones increases uncertainty and can bias results, even when the foreground plant data are plausibility-checked and the overall data quality is considered good. However, the number of such datasets among background data is very limited and minimal.

3.2 System boundary

This EPD covers the life cycle assessment of the Lindner Logic 100 Timber Partition Wall System for the life cycle stages '**cradle to gate with options**,' including modules A1–A3, C, and D, with modules A4 and A5 reported as additional modules. The description of the modules is as follows:

- Module A1: Extraction and processing of raw materials. This includes steel (sub-structure/connection profiles), wood-based materials (wall panel and surface layer), plastics and rubber (e.g., sealants), gypsum-based materials (sound insulation and fire protection), and mineral wool.
- Module A2: Transport of raw materials and supplies from each supplier to the production plant. The real distances to the specific factory gate (Arnstorf or Ostrov) are considered per material.
- Module A3: Manufacturing at the plants, including energy consumption (electricity and thermal energy) at both sites; no water use is required. It also includes handling/disposal of production residues and the production of packaging materials. Residues are landfilled, incinerated, or recycled depending on the material type, with 50 km assumed to the waste facility. Because these residues do not reach end-of-waste status in A3, no credits are taken in Module D for energy recovered from the incineration of production residues. Packaging includes reusable wooden pallets, chipboard, PE-based plastics, and cardboard; for pallets, for example, 95 % reuse and 5 % disposal/incineration are assumed.
- Module A4: Transport of the product from the factory gate to the construction site; 100 km is assumed.
- Module A5: Installation on site and disposal of packaging.
- Modules B1–B2: Not declared.
- Modules B3–B5: Not relevant.
- Modules B6–B7: Not declared.
- Module C1: Dismantling/removal from the building. Due to the modular design, no destructive demolition is needed, and there are no significant environmental impacts.
- Module C2: Transport of dismantled components to the waste treatment/disposal facility; 50 km is assumed.
- Module C3: Recycling of metal components and incineration (with energy recovery) of wood- and plastic-based components. Reused or repurposed components are also reported in C3.
  Module C4: Final disposal of non-reusable or non-repurposable materials/components, including landfilling of gypsum-based components and mineral wool.
- Module D: Benefits and loads beyond the system boundary (credits), such as from recycling (steel), energy recovery (wood/plastics), and partial reuse or repurposing. Credits are only given for materials/energy that have reached end-of-waste status, and they vary by end-of-life scenario.

3.3 Estimates and assumptions

Assumptions and estimates were required during the LCA modelling process, mainly due to the use of proxy datasets in Modules A1–A3 (e.g. for veneer, adhesives, and surface coatings), packaging modelling in Modules A3/A5 (including the reuse and disposal shares of reusable wooden pallets), assumed transport distances in Modules A4 and C2 (100 km to the construction site and 50 km to waste treatment facilities, with additional 100 km transport assumed for reuse and repurposing scenarios), and the definition of end-of-life scenarios in Modules C and D (conventional disposal, reuse, and repurposing scenarios reflecting modular design and take-back practices).

All assumptions were defined conservatively, are consistent with common LCA practice and the requirements of EN 15804. Detailed explanations, technical justifications, data sources, and supporting evidence for the validity of these assumptions are fully documented in the LCA background report.

3.4 Cut-off criteria

Data for the manufacturing process were collected by the life cycle assessor, with foreground data referring to the year 2023. No mandatory life cycle stages or relevant processes were omitted, and all known and quantifiable inputs and outputs were considered using available LCI datasets from the LCA FE databases accessible to the LCA provider.
The Logic 100 Timber system comprises a wide range of product variants. To simplify the EPD and limit the number of variants covered, environmental impact calculations for different natural veneer surface layers (spruce, beech, and oak) were not modelled separately. Instead, it was assumed that the environmental impacts of the HPL surface variant are representative for all veneer options. This approach is conservative, as the HPL variant requires a higher material input (2.24 kg/m²) and exhibits higher environmental impacts per kilogram compared to natural wood veneers, which require lower material quantities.
Based on this worst-case approach, product variants were grouped in two stages: first by selecting the most impactful surface layer, and subsequently by grouping products with similar technical properties and environmental profiles under a single EPD.
No additional cut-off criteria were applied to input or output flows.

3.5 Background data

The background data is primarily drawn from the *GaBi LCA FE database, Sphera MLC (fka GaBi) CUP 2023.02*, which provides either technology-specific data or average data relevant to the analysis.

3.6 Data quality

The data quality of the foreground data in the life cycle inventory (LCI) is rigorously evaluated to meet the standards outlined in ***EN 15804+A2*** and ***EN ISO 14044:2006***. The data sets align with the current ILCD format, though not all materials could be represented by complete life cycle inventories.
The collected data is subsequently processed and analysed using *GaBi* software, with consistent application of allocations and methodological principles throughout the software model. This approach ensures that representativeness, completeness, data reliability, and consistency are all satisfactorily achieved.

- **Technological coverage:**
  The data used in the LCA reflect the specific technologies employed at both the German and the Czech Republic production facilities. Where technology-specific data were not available, approximations were made using the closest representative datasets from the GaBi database to maintain technological representativeness.
- **Time-related coverage:**
  Primary data collected from the production facilities represent the most recent production conditions and refer to the year 2023. The reference years of the generic background datasets from the GaBi database are consistent with the project timeframe, ensuring appropriate temporal representativeness.
- **Geographical coverage:**
  Foreground data were collected from production facilities in Germany and the Czech Republic. Background data were sourced from the GaBi database, which provides geographically relevant datasets for both Germany and the Czech Republicand under module A1 to A3, depending on the geographical origin of the supplier. For the construction site and the end of life, European or global processes have been used in the software, ensuring the regional and European environmental conditions are appropriately reflected.
- **Completeness:**
  All relevant processes and material and energy flows associated with the production system were considered in the LCA. Data were systematically categorised into energy, raw materials, packaging materials and waste.

3.7 Period under review

Production data for this assessment were collected at the Lindner plants in Arnstorf, Germany, and Ostrov Czech Republic for the year 2023.

3.8 Geographic Representativeness

Land or region, in which the declared product system is manufactured, used or handled at the end of the product´s lifespan: Europe

3.9 Allocation

The production plants at Arnstorf and Ostrova produce not only the intended product, Logic 100 Timber, and its components, but also different co-products. In accordance with [***EN 15804+A2**, Section 6.4.3.2*], in multi-product facilities where allocation cannot be avoided, allocation may be based on a physical relationship between materials or products and be applied according to their share of total production.
In this case, to determine the specific environmental impacts related to Logic 100 Timber, the necessary information, including energy and waste data, has been collected from both production plants. The actual amount of raw material inputs entering the factory has also been collected. The list of materials has been checked, and materials that undergo processes like cutting, welding, or bending and etc., have been identified. Energy and waste have been allocated to these materials based on their **mass quantities**. Subsequently, allocations have been made for the functional unit based on the amount of each material it includes.

3.10 Comparability

Basically, a comparison or an evaluation of EPD data is only possible if all the data sets to be compared were created according to *EN 15804* and the building context, respectively the product-specific characteristics of performance, are taken into account. The background data is primarily drawn from the *GaBi LCA FE database, Sphera MLC (fka GaBi) CUP 2023.02*, which provides either technology-specific data or average data relevant to the analysis.

4. LCA: Scenarios and additional technical information

Characteristic product properties of biogenic carbon

The biogenic carbon content quantifies the amount of biogenic carbon present in Logic 100 Timber variants as it leaves the factory gate and in its accompanying packaging. For the modules which include the biobased material or a process like incineration of this biobased material, these values have been calculated. The biogenic carbon content of the fuel has been balanced in the transportation dataset. The biogenic carbon content of the declared product and the corresponding packaging is presented in the table below.

Information on describing the biogenic carbon content at factory gate

Name	Value	Unit
Biogenic carbon content in product	11.76	kg C
Biogenic carbon content in accompanying packaging	0.22	kg C

Note: 1 kg of biogenic carbon is equivalent to 44/12 kg of CO_2.

The following technical information is the basis for the declared modules. A5 is not declared, including the disposal of the packaging material on the construction site, since the amount of packaging materials included in the LCA calculations is equal to the amount of packaging material to be disposed of under A5.

Transport to the building site (A4)

Name	Value	Unit
Fuel amount	0.105	kg/m2
Transport distance	100	km
Capacity utilisation (including empty runs)	61	%
Product weight to be transferred, inc. packaging	45.2	kg/m2

Installation into the building (A5)

Module A5 includes the installation phase, as well as the disposal of packaging materials. The individual components of the Logic 100 timber partitions are assembled manually on the construction site, and there are no relevant environmental impacts, nor indoor air pollution incurred by the installation of the product.
Regarding the disposal of packaging materials, the exact amount of packaging materials entered the system under A3 will be disposed of under A5. 95% of wooden pallets are reusable (no environmental burden), while the remaining 5% are incinerated. All other packaging materials (cardboard and plastic-based) are also incinerated under A5. The thermal energy recovered from incineration is reported under module D (no MER). For packaging cardboard, the SM content (87.4%) has been subtracted from module D credits, as no benefits are associated with energy extracted from secondary material content.

Use stage (B)

Module B is not included in the system boundary of this EPD. Modules B1, B2, B6, and B7 are not declared, while Modules B3, B4, and B5 are not relevant.

The useful life of non-load-bearing walls is ≥ 50 years (according to the *BBSR* table, code no. 342.411, as of 02/2017, published by the Federal Institute for Research on Building, Urban Affairs and Spatial Development). If used properly, no costs for maintenance, repair, or replacement are expected during this period.

End of life (C1-C4)

The C1-C4 modules cover the disposal phase for the Lindner Logic 100 Timber Partition Wall System, addressing three different end-of-life (EoL) scenarios for the product, including:

- Scenario 0 - Conventional Disposal
- Scenario 1 - Reuse
- Scenario 2- Repurpose

Module C1: This includes the dismantling of the partition walls from the building, which, due to the modular design of the product, requires no destructive demolition or the use of power or electricity and does not impose any significant environmental impact.

Module C2: Transferring the components of the product to the relevant waste disposal facility, with an assumed distance of 50 km.

Module C3: As detailed below:

Scenario 0 - Conventional Disposal

- Steel-based components: **Recycling**
- Plastics and sealants: Including acrylic edgebands, EPDM, PE, and TPE, **Incineration** and thermal recovery (Electricity and heat extraction)
- Wood-based components: Including the chipboard and the surface layer veneer, **Incineration** and thermal recovery (Electricity and heat extraction)
- Gypsum-based components: Including GFP, GKB, GKF, and GKFI, **Landfill**
- Mineral wool: **Landfill**

Scenario 1 - Reuse

- Steel-based components: Including system stand ceiling bracket, system stand height adjustable bracket and all floor/ceiling/wall connection profiles, **Recycling**
- Plastics and sealants: Including EPDM, PE, and TPE, **Incineration** and thermal recovery (Electricity and heat extraction)
- The rest of the product: **ReUse** and booked as CRU under C3 (components for reuse)

Scenario 2- Repurpose

- Steel-based components: Including system stand ceiling bracket, system stand height adjustable bracket and all floor/ceiling/wall connection profiles, Recycling
- Plastics and sealants: Including EPDM, PE, and TPE, **Incineration** and thermal recovery (Electricity and heat extraction)
- Gypsum-based components: Including GFP, GKB, GKF, and GKFI, **Landfill**
- Mineral wool: **Landfill**
- The wall panel with all components: **ReUse** and booked as CRU under C3 (components for reuse)

The exact quantity of the materials and the disposal method for the declared unit have been shown in the table below.

Name	Value	Unit
Collected separately waste type equal to the weight of product	44.51	kg
Reuse - Sc. 0	-	kg
Recycling - Sc. 0	5.64	kg
Energy recovery - Sc. 0	27.12	kg
Landfilling - Sc. 0	11.75	kg
Reuse - Sc. 1	42.97	kg
Recycling - Sc. 1	1.25	kg
Energy recovery - Sc. 1	0.23	kg
Landfilling - Sc. 1	0.066	kg
Reuse - Sc. 2	37.82	kg
Recycling - Sc. 2	5.61	kg
Energy recovery - Sc. 2	0.20	kg
Landfilling - Sc. 2	0.87	kg

Reuse, recovery and/or recycling potentials (D), relevant scenario information

Module D credits are calculated based on avoided burdens from the reuse and recycling of materials or components, as well as from electrical and thermal energy recovery resulting from the incineration of materials and packaging, both at the end of life and under module A5. For Module D calculations, the share of secondary material is excluded before accounting for recycling credits. For incineration processes, the water content is excluded before calculating energy recovery credits.

Name	Value	Unit
Material for reuse (CRU) - Sc. 0	0.23	kg
Material for recycling (MFR) - Sc. 0	4.72	kg
Material for energy recovery (MER) - Sc. 0	27.12	kg
Material for reuse (CRU) - Sc. 1	43.20	kg
Material for recycling (MFR) - Sc. 1	1.04	kg
Material for energy recovery (MER) - Sc. 1	0.23	kg
Material for reuse (CRU) - Sc. 2	38.05	kg
Material for recycling (MFR) - Sc. 2	4.70	kg
Material for energy recovery (MER) - Sc. 2	0.20	kg

5. LCA: Results

Information on the environmental impacts is established using the characterisation factors according to /CML, as published in April 2013/. Long-term emissions are not taken into consideration. The characterisation factors applied correspond with the requirements of Annex C in ***DIN EN 15804***.
Scenario 0 - Conventional Disposal; represented by C2, C3, C4, D
Scenario 1 - Reuse; represented by C2/1, C3/1, C4/1, D/1
Scenario 2 - Repurpose; represented by C2/2, C3/2, C4/2, D/2

Note: The following emission factors have been used for the calculation of CO_2 results under module A3, extracted from the Sphera 2023.02 database.

DE: Reststrommix Sphera = 0.6660 Kg CO_2 e/KWh
DE: Thermische Energie aus Erdgas Sphera = 0.2328 CO_2 e/KWh
CZ: Strom Mix Sphera = 0.5567 CO_2 e/KWh
CZ: Hackschnitzelkessel 120-400 kW (*EN15804 B6*) Sphera = 0.0059 CO_2 e/KWh

DESCRIPTION OF THE SYSTEM BOUNDARY (X = INCLUDED IN LCA; MND = MODULE OR INDICATOR NOT DECLARED; MNR = MODULE NOT RELEVANT)

Product stage			Construction process stage		Use stage							End of life stage				Benefits and loads beyond the system boundaries
Raw material supply	Transport	Manufacturing	Transport from the gate to the site	Assembly	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
A1	**A2**	**A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
X	X	X	X	X	MND	MND	MNR	MNR	MNR	MND	MND	X	X	X	X	X

RESULTS OF THE LCA - ENVIRONMENTAL IMPACT according to EN 15804+A2: 1 m² Lindner Logic 100 Timber partition, 47 and 49 dB Sound Proffing Variants

Parameter	Unit	A1-A3	A4	A5	C1	C2	C2/1	C2/2	C3	C3/1	C3/2	C4	C4/1	C4/2	D	D/1	D/2
GWP-total	kg CO_2 eq	-6.87E+00	3.71E-01	1.44E+00	0	1.81E-01	3.55E-01	3.35E-01	4.43E+01	4.37E+01	4.29E+01	8.71E-01	0	7.09E-01	-2.83E+01	-8.08E+01	-7.67E+01
GWP-fossil	kg CO_2 eq	3.7E+01	3.67E-01	6.39E-01	0	1.79E-01	3.52E-01	3.32E-01	1.87E+00	5.58E-01	5.15E-01	1.83E-01	0	2.16E-02	-2.83E+01	-3.66E+01	-3.27E+01
GWP-biogenic	kg CO_2 eq	-4.39E+01	1.65E-03	8.02E-01	0	5.1E-04	1E-03	9.44E-04	4.24E+01	4.31E+01	4.24E+01	6.87E-01	0	6.87E-01	-4.69E-02	-4.41E+01	-4.4E+01
GWP-luluc	kg CO_2 eq	2.76E-02	2.21E-03	2.11E-04	0	1.09E-03	2.13E-03	2.01E-03	1.3E-05	1.27E-05	8.6E-06	5.49E-04	0	3.85E-05	-5.67E-03	-2.8E-02	-2.74E-02
ODP	kg CFC11 eq	2.65E-08	9.1E-14	2.85E-13	0	4.48E-14	8.79E-14	8.29E-14	6E-08	1.83E-13	1.32E-13	4.65E-13	0	4.3E-14	-6.62E-11	-9.27E-08	-9.27E-08
AP	mol H^+ eq	9.88E-02	4.91E-04	2.27E-04	0	2.42E-04	4.74E-04	4.47E-04	2.45E-02	8.83E-05	7.74E-05	1.26E-03	0	1E-04	-4.39E-02	-1.1E-01	-9.77E-02
EP-freshwater	kg P eq	2.83E-04	8.7E-07	1.49E-07	0	4.28E-07	8.39E-07	7.92E-07	3.64E-07	5.28E-08	3.91E-08	3.6E-07	0	2.84E-08	-2.16E-05	-2.98E-04	-2.92E-04
EP-marine	kg N eq	2.79E-02	1.84E-04	6.3E-05	0	9.04E-05	1.77E-04	1.67E-04	1.06E-02	2.09E-05	1.76E-05	3.26E-04	0	2.56E-05	-1.19E-02	-2.94E-02	-2.75E-02
EP-terrestrial	mol N eq	2.78E-01	2.17E-03	1.02E-03	0	1.07E-03	2.1E-03	1.98E-03	1.24E-01	4.12E-04	3.64E-04	3.59E-03	0	2.81E-04	-1.29E-01	-2.94E-01	-2.74E-01
POCP	kg NMVOC eq	8.79E-02	4.36E-04	1.7E-04	0	2.15E-04	4.21E-04	3.97E-04	2.73E-02	5.72E-05	4.78E-05	9.86E-04	0	7.84E-05	-3.67E-02	-9.25E-02	-8.5E-02
ADPE	kg Sb eq	2.42E-04	2.67E-08	4.51E-09	0	1.31E-08	2.58E-08	2.43E-08	-1.71E-06	1.45E-09	1.04E-09	8.36E-09	0	7.44E-10	-1.07E-06	-1.57E-04	-6.69E-06
ADPF	MJ	6.19E+02	5.02E+00	8.63E-01	0	2.47E+00	4.84E+00	4.57E+00	1.83E+01	2.71E-01	1.96E-01	2.5E+00	0	3.11E-01	-3.91E+02	-6.36E+02	-5.69E+02
WDP	m^3 world eq deprived	9.14E+00	1.94E-03	9.67E-02	0	9.53E-04	1.87E-03	1.76E-03	4.99E+00	5.98E-02	5.57E-02	1.86E-02	0	6.75E-04	-1.35E+00	-1.08E+01	-1.07E+01

GWP = Global warming potential; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential of land and water; EP = Eutrophication potential; POCP = Formation potential of tropospheric ozone photochemical oxidants; ADPE = Abiotic depletion potential for non-fossil resources; ADPF = Abiotic depletion potential for fossil resources; WDP = Water (user) deprivation potential)

RESULTS OF THE LCA - INDICATORS TO DESCRIBE RESOURCE USE according to EN 15804+A2: 1 m² Lindner Logic 100 Timber partition, 47 and 49 dB Sound Proffing Variants

Parameter	Unit	A1-A3	A4	A5	C1	C2	C2/1	C2/2	C3	C3/1	C3/2	C4	C4/1	C4/2	D	D/1	D/2
PERE	MJ	6.73E	3.36E-	3.84E	0	1.66E-	3.25E-	3.06E-	3.96E	9.07E-	6.52E-	1.27E	0	3.57E-	-5.38E	-7.06E	-6.85E

Lindner

		+02	01	+00		01	01	01	+02	02	02	+01		02	+01	+02	+02
PERM	MJ	0	0	-3.13E+00	0	0	0	0	-3.96E+02	-4.08E+02	-4.01E+02	-1.23E+01	0	-7.08E+00	0	0	0
PERT	MJ	1.08E+03	3.36E-01	7.13E-01	0	1.66E-01	3.25E-01	3.06E-01	5.34E-01	-4.08E+02	-4.01E+02	3.93E-01	0	-7.04E+00	-5.38E+01	-7.06E+02	-6.85E+02
PENRE	MJ	6.2E+02	5.03E+00	1.31E+01	0	2.47E+00	4.85E+00	4.58E+00	5.29E+01	3.48E+01	8.57E+00	2.5E+00	0	3.11E-01	-3.92E+02	-6.36E+02	-5.7E+02
PENRM	MJ	0	0	-1.04E+01	0	0	0	0	-3.45E+01	-3.45E+01	-3.45E+01	0	0	0	0	0	0
PENRT	MJ	6.65E+02	5.03E+00	2.71E+00	0	2.47E+00	4.85E+00	4.58E+00	1.83E+01	2.71E-01	-2.59E+01	2.5E+00	0	3.11E-01	-3.92E+02	-6.36E+02	-5.7E+02
SM	kg	1.16E+00	0	1.81E-01	0	0	0	0	0	0	0	0	0	0	0	7.4E-01	5.64E-03
RSF	MJ	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NRSF	MJ	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
FW	m^3	9.55E-01	2.99E-04	2.33E-03	0	1.47E-04	2.89E-04	2.72E-04	1.16E-01	1.42E-03	1.31E-03	5.8E-04	0	2.88E-05	-6.19E-02	-9.89E-01	-2.92E-01

PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy resources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water

RESULTS OF THE LCA – WASTE CATEGORIES AND OUTPUT FLOWS according to EN 15804+A2: 1 m² Lindner Logic 100 Timber partition, 47 and 49 dB Sound Proffing Variants

Parameter	Unit	A1-A3	A4	A5	C1	C2	C2/1	C2/2	C3	C3/1	C3/2	C4	C4/1	C4/2	D	D/1	D/2
HWD	kg	3.81E-06	8.47E-12	-4.01E-12	0	4.17E-12	8.18E-12	7.71E-12	-4.61E-12	-5.51E-12	-3.31E-12	6.65E-11	0	1.93E-11	-2.59E-08	-2.73E-06	-8.34E-07
NHWD	kg	1.42E+00	7.52E-04	6.88E-02	0	3.7E-04	7.26E-04	6.85E-04	5.48E-02	5.37E-02	3.48E-02	1.18E+01	0	8.21E-01	-3.03E-01	-1.69E+00	-1.63E+00
RWD	kg	1.41E-02	6.61E-06	1.48E-05	0	3.26E-06	6.38E-06	6.02E-06	9.81E-04	9.26E-06	6.67E-06	2.82E-05	0	3.58E-06	-1.63E-02	-1.18E-02	-1.01E-02
CRU	kg	0	0	2.31E-01	0	0	0	0	0	4.3E+01	3.78E+01	0	0	0	0	4.7E+01	4.17E+01
MFR	kg	2.51E-01	0	0	0	0	0	0	5.64E+00	1.25E+00	5.61E+00	0	0	0	0	0	0
MER	kg	0	0	0	0	0	0	0	2.71E+01	2.25E-01	2.04E-01	0	0	0	0	0	0
EEE	MJ	0	0	1.35E+00	0	0	0	0	5.34E+01	1.1E+00	1.05E+00	0	0	0	1.1E+00	1.1E+00	1.05E+00
EET	MJ	0	0	3.12E+00	0	0	0	0	1.62E+02	2.51E+00	2.4E+00	0	0	0	2.51E+00	2.51E+00	2.4E+00

HWD = Hazardous waste disposed; NHWD = Non-hazardous waste disposed; RWD = Radioactive waste disposed; CRU = Components for re-use; MFR = Materials for recycling; MER = Materials for energy recovery; EEE = Exported electrical energy; EET = Exported thermal energy

RESULTS OF THE LCA – additional impact categories according to EN 15804+A2-optional: 1 m² Lindner Logic 100 Timber partition, 47 and 49 dB Sound Proffing Variants

Parameter	Unit	A1-A3	A4	A5	C1	C2	C2/1	C2/2	C3	C3/1	C3/2	C4	C4/1	C4/2	D	D/1	D/2
PM	Disease incidence	2.05E-06	4.03E-09	1.93E-09	0	1.98E-09	3.89E-09	3.67E-09	8.74E-08	1.13E-09	8.96E-10	1.55E-08	0	1.16E-09	-5.29E-07	-2.15E-06	-1.35E-06
IR	kBq U235 eq	1.78E+00	7.08E-04	1.58E-03	0	3.49E-04	6.84E-04	6.45E-04	1.68E-02	9.78E-04	7.04E-04	3.28E-03	0	4.89E-04	-2.8E+00	-1.8E+00	-1.52E+00
ETP-fw	CTUe	1.97E+02	3.66E+00	4.81E-01	0	1.8E+00	3.53E+00	3.33E+00	2.3E+00	1.03E-01	7.52E-02	1.3E+00	0	1.18E-01	-4.59E+01	-1.91E+02	-1.85E+02
HTP-c	CTUh	3.93E-08	7.3E-11	1.88E-11	0	3.59E-11	7.05E-11	6.65E-11	6.07E-10	7.37E-12	5.55E-12	2E-10	0	1.61E-11	-2.06E-08	-4.57E-08	-5.11E-08
HTP-nc	CTUh	4.76E-07	3.06E-09	1.06E-09	0	1.51E-09	2.95E-09	2.79E-09	4.55E-08	3.68E-10	2.36E-10	2.1E-08	0	1.62E-09	-7.29E-08	-4.45E-07	-3.92E-07
SQP	SQP	1.83E+03	1.79E+00	2.87E-01	0	8.79E-01	1.72E+00	1.63E+00	9.54E-02	8.3E-02	6.13E-02	5.97E-01	0	4.57E-02	-3.67E+01	-1.91E+03	-1.89E+03

PM = Potential incidence of disease due to PM emissions; IR = Potential Human exposure efficiency relative to U235; ETP-fw = Potential comparative Toxic Unit for ecosystems; HTP-c = Potential comparative Toxic Unit for humans (cancerogenic); HTP-nc = Potential comparative Toxic Unit for humans (not cancerogenic); SQP = Potential soil quality index

Disclaimer 1 – for the indicator 'Potential Human exposure efficiency relative to U235'. This impact category deals mainly with the eventual impact of low-dose ionizing radiation on human health of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure or radioactive waste disposal in underground facilities. Potential ionizing radiation from the soil, radon and from some construction materials is also not measured by this indicator.

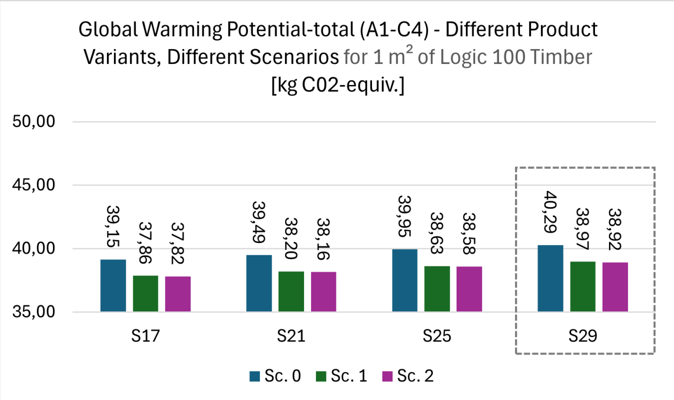
Disclaimer 2 – for the indicators 'abiotic depletion potential for non-fossil resources', 'abiotic depletion potential for fossil resources', 'water (user) deprivation potential, deprivation-weighted water consumption', 'potential comparative toxic unit for ecosystems', 'potential comparative toxic unit for humans – cancerogenic', 'Potential comparative toxic unit for humans - not cancerogenic', 'potential soil quality index'. The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high as there is limited experience with the indicator.

6. LCA: Interpretation

This EPD covers the Logic 100 Timber, 47 and 49 dB Soundproofing Partition Wall Systems, including four specific product variants. The representative declared unit is the Logic 100 Timber 49 dB, with an 82 mm substructure and HPL veneer surface cover, weighing 33.64 kg/m², which is the heaviest product in the group.

a. Global Warming Potential Deviations in the Product Family

All four product variants are sound-insulation timber partitions labelled S17, S21, S25, and S29 for easier reference in the LCA studies. The maximum deviation in GWP-total across all life cycle stages (A1-C4) is 2.85 %, as shown in the diagram below. S29, having the highest weight, also has the greatest environmental impact, with a GWP-total of 40.29 kg CO_2-equivalent in Scenario 0, 38.97 kg CO_2-equivalent in Scenario 1, and 38.92 kg CO_2-equivalent in Scenario 2.



b. Global Warming Potential over Life Cycle Stages and End of Life Scenarios (only the declared unit)

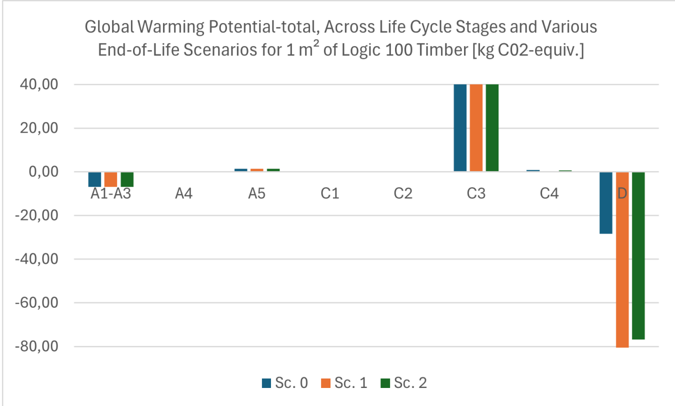
The assessment of Global Warming Potential (GWP) across different life cycle stages and scenarios reveals during the production phase (A1-A3), the environmental impact remains stable across all scenarios, marked by a consistent value of -6.87 kg CO_2-equivalent.

Similarly, the transportation and construction phases, represented by modules A4 and A5, show uniform GWP values across different scenarios. With impacts of 0.37 kg CO_2-equivalent for transportation and 1.44 kg CO_2-equivalent for construction, these stages highlight the standard environmental costs associated with logistics and assembly. However, variability emerges in the end-of-life modules. In Sc. 0, where conventional disposal is practised, the GWP peaks at 44.29 kg CO_2-equivalent in module C3. This high GWP is due to the release of sequestered and bound biogenic carbon content during the incineration of chipboard. Additionally, the incineration process contributes to a high GWP-fossil impact for both the chipboard and plastic components.

In Sc. 1 and 2, which involve reusing and repurposing major parts of the product instead of incineration, a lower GWP-total under module C3 is expected. However, due to the product's high biogenic carbon content, stemming from its timber nature, this amount should still be booked under module C3 as GWP-biogen. Thus, even though incineration is limited to minor plastic components and not the chipboard in these scenarios, the result is a lower GWP-fossil but still a high GWP-biogen and overall GWP-total. Also in Sc. 2, some parts of the product, which are made out of gypsum fibre board, get landfilled under C4 and the relevant biogenic carbon is booked under this module.

S29																
Indicator	A1-A3	A4	A5	C1	C2	C2/1	C2.2	C3	C3/1	C3/2	C4	C4/1	C4/2	D	D/1	D/2
GWP-total	-6,87	0,37	1,44	0,00	0,18	0,35	0,33	44,29	43,67	42,94	0,87	0,00	0,71	-28,34	-80,77	-76,69
GWP-fossil	36,34	0,37	0,64	0,00	0,18	0,35	0,33	1,87	0,56	0,51	0,18	0,00	0,02	-28,28	-36,63	-32,70
GWP-biogenic	-43,17	0,00	0,80	0,00	0,00	0,00	0,00	42,43	43,11	42,43	0,69	0,00	0,69	-0,05	-44,12	-43,97
GWP-luluc	0,03	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	-0,01	-0,03	-0,03

Module D explores the end-of-life benefits extensively. Sc. 0 offers limited environmental advantages with a GWP-total of -28.34 kg CO_2-equivalent, compared to Sc. 1 and Sc. 2, achieving a GWP of -80.77 kg CO_2-equivalent and -76.69 kg CO_2-equivalent, respectively, which amplify the benefits of reuse and repurpose beyond the system boundary for the next life cycle of the product. These values indicated optimising emission offsets.



c. Relative Contribution across Environmental Indicators (only the declared unit)

The analysis of relative contributions across core environmental indicators for Sc. 0 highlights significant insights into module impacts.

The Production Phase (A1-A3) emerges as a dominant contributor across nearly all impact categories. It accounts for the majority of fossil emissions (92 % for GWP-fossil) and significantly contributes to natural resource depletion (95 % for ADPF). It also records the highest contribution in abiotic depletion potential of elements (ADPE) and freshwater eutrophication potential (EP-freshwater), both at 99 %. This highlights extensive resource use and potential pollution during this phase. However, it also exhibits carbon sequestration benefits, with negative impacts in total global warming potential (-13 % for GWP-total) and biogenic carbon emissions (-50 % for GWP-biogenic).

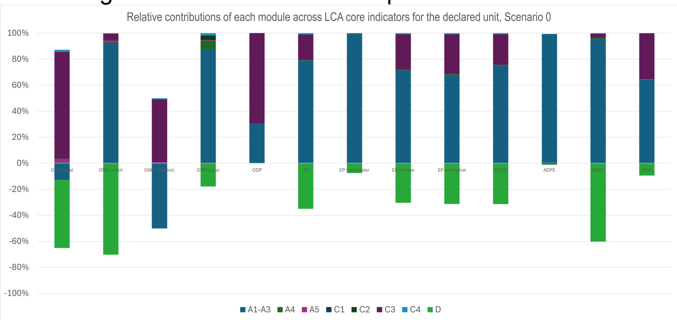
Transportation (A4) influences impact categories minimally, showing only slight contributions to fossil emissions (1 %) and land use change-related carbon emissions (7 % for GWP-luluc).

Construction (A5) contributes slightly to fossil emissions (3 %) and water usage (1 % for WDP).

Deconstruction and Waste Processing (C1, C2, C3) have varied impacts. Waste processing (C3) significantly affects total global warming potential (83 % for GWP-total) and ozone depletion potential (69 % for ODP) due to emissions during treatment. It also impacts marine (27 % for EP-marine) and terrestrial eutrophication potential (30 % for EP-terrestrial).

Final Disposal (C4) has minimal contributions across all categories, indicating an insignificant amount of landfilled materials.

Beyond System Boundaries (Module D) demonstrates substantial environmental benefits, notably reducing fossil emissions (-70 % for GWP-fossil) and resource depletion (-60 % for ADPF). These reductions underscore the advantages of material recycling, such as steel, and energy recovery through burning plastic-based packaging materials, effectively minimising carbon and resource footprints.



7. Requisite evidence

The test report 392-2018-00188001_A_DE, dated July 27, 2018, is available for the Logic 100 Timber partitions. The specimen was produced on 08/05/2018 and collected for testing on 14/05/2018.
The testing was conducted by Eurofins Product Testing A/S, located at Smedeskovvej 38, DK-8464 Galten, Denmark.
The tested product is suitable for indoor use and fulfils the requirements of Indoor Air Comfort GOLD® in accordance with the Indoor Air Comfort GOLD 6.0 regulation (February 2017).

7.1 VOC emissions

AgBB overview of results, VOC emissions (28 days [µg/m³])

Name	Value	Unit
TVOC (C6 - C16)	< 5	µg/m3
Sum SVOC (C16 - C22)	< 5	µg/m3
R (dimensionless)	0.091	-
VOC without NIK	< 5	µg/m3
Carcinogenic Substances	< 1	µg/m3

7.2 Aldehydes and Formaldehyde

The existence of Formaldehyde as well as other aldehydes are reported as follows after 28 days:

AgBB overview of results, Aldehydes and Formaldehyde (28 days [µg/m³])

Name	Value	Unit
Formaldehyde	9.1	µg/m³
Acetaldehyde	< 3	µg/m³
Propionaldehyd	< 3	µg/m³
Butyraldehyde	< 3	µg/m³
2-butenal	< 5	µg/m³
Glutaraldehyd	< 5	µg/m³

8. References

IBU 2021
Institut Bauen und Umwelt e.V.: General Instructions for the EPD programme of Institut Bauen und Umwelt e.V., Version 2.0, Berlin: Institut Bauen und Umwelt e.V., 2021
www.ibu-epd.com

LCA software
Sphera LCA for Experts (Former GaBi) version 10.7.1.28

LCA database
Sphera MLC (fka GaBi) CUP 2023.02 database

<u>Standards:</u>

EN 15804:2012+A1 2013
Sustainability of construction works — Environmental Product Declarations — Core rules for the product category of construction products.

EN 15804:2012+A2:2019+AC:2021
Sustainability of construction works — Environmental Product Declarations — Core rules for the product category of construction products.

EN ISO 14025:2011
Environmental labels and declarations — Type III environmental declarations — Principles and procedures.
Further References

EN ISO 14044:2006+A2:2020
Environmental management — Life cycle assessment — Requirements and guidelines.

EN ISO 9001:2015
Quality management systems — Requirements.

EN ISO 14001:2015
Environmental management systems — Requirements with guidance for use.

EN ISO 50001:2018
Energy management systems — Requirements with guidance for use.

DIN 4103-1
Internal non-load-bearing partitions — Requirements and verification.

DIN EN ISO 10140-2
Acoustics — Laboratory measurement of sound insulation of building elements — Part 2: Measurement of airborne sound insulation.

DIN EN ISO 10848-2
Acoustics — Laboratory measurement of the flanking transmission of airborne sound between adjoining rooms — Part 2.

EN 13501-1
Fire classification of construction products and building elements — Part 1: Classification using data from reaction to fire tests.

<u>Other References:</u>

PCR part A
Calculation Rules for the Life Cycle Assessment and Requirements on the Project Report according to EN 15804+A2:2019, Version 1.4.

PCR part B PCR
Guidance Texts for Building-Related Products and Services — Requirements on the PD for System Floors, Version 10, last amended 30.04.2024.

REACH Regulation
European Parliament and Council. (2006). Regulation (EC) No1907/2006 concerning the Registration, Evaluation, Authorisation, and Restriction of Chemicals (REACH). Official Journal of the European Union, L396, 1-849.

BBSR Table (code no. 342.411, as of 02/2017) Federal Institute for Research on Building, Urban Affairs and Spatial Development (BBSR) (2017).
Service life table for building components, Code No. 342.411, as of 02/2017.

Hazardous Substances Ordinance (Germany)
Annex IV, No. 22.

Chemicals Prohibition Ordinance (Germany)
Annex to § 1, No. 23.

German Ordinance on Industrial Safety and Health (Betriebssicherheitsverordnung)

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